

COMP3013 2024 Fall

Assignment 1

Note: there is no requirement on ER diagram drawing. You are allowed to draw ER diagrams using any tool. But you need to make sure that your drawing is clear enough. TAs have rights to remove marks if your graph unreadable. Please do not try to challenge us.

For the submission, please pack your answers to Q2 and Q3 to **A SINGLE PDF FILE**. Rename the PDF file as COMP3013_23F_A1_XXXX, where XXXX is your student ID. Answers of Q1 are submitted via **iSpace Feedback**. Submissions which do not following the guideline may **not be marked**.

Q1. The schema of a database is given as follows. Keys are underlined.

- *book* = (*bID*, *bname*, *year*, *author*)
- *reviewer* = (*rID*, *rname*)
- *rating* = (*rID*, *bID*, *score*, *date*) // *score* is an integer and $1 \leq score \leq 5$
- *comment* = (*rID*, *bID*, *content*)

Write a query for each following question. Please submit your answers of this question via **iSpace Feedback**. (7 marks for each)

- a) Find the name of books which are written by Kleene (author).

```
SELECT BNAME
FROM BOOK
WHERE AUTHOR = 'KLEENE'
```
- b) Find the name of books which have received a rating score less than 3.

```
SELECT BNAME
FROM BOOK, RATING
WHERE BOOK.BID = RATING.BID
AND SCORE < 3
```
- c) Find the name of reviewers who have made a comment to book “Quantum Finance” (book name).

```
SELECT RNAME
FROM BOOK, REVIEWER, COMMENT
WHERE BOOK.BID = COMMENT.BID
AND REVIEWER.RID = COMMENT.RID
AND BNAME = 'QUANTUM FINANCE'
```
- d) Find the name of reviewers who have written some book.

```
SELECT RNAME
FROM REVIEWER, BOOK
WHERE REVIEWER.RNAME = BOOK.AUTHOR
```

- e) Find the authors who have written multiple books.

```
SELECT AUTHOR
FROM BOOK AS B1, BOOK AS B2
WHERE B1.AUTHOR = B2.AUTHOR
AND B1.BID <> B2.BID
```

- f) Find the name of books which have been rated at both score 1 and score 5.

```
SELECT BNAME
FROM BOOK, RATING
WHERE BOOK.BID = RATING.BID
AND SCORE = 1
INTERSET
SELECT BNAME
FROM BOOK, RATING
WHERE BOOK.BID = RATING.BID
AND SCORE = 5
```

Select b1.bname

From book as b1, book as b2, rating as r1, rating as r2

Where b1.bid = r1.bid and r1.score=1

And b2.bid = r2.rid and r2.score = 5

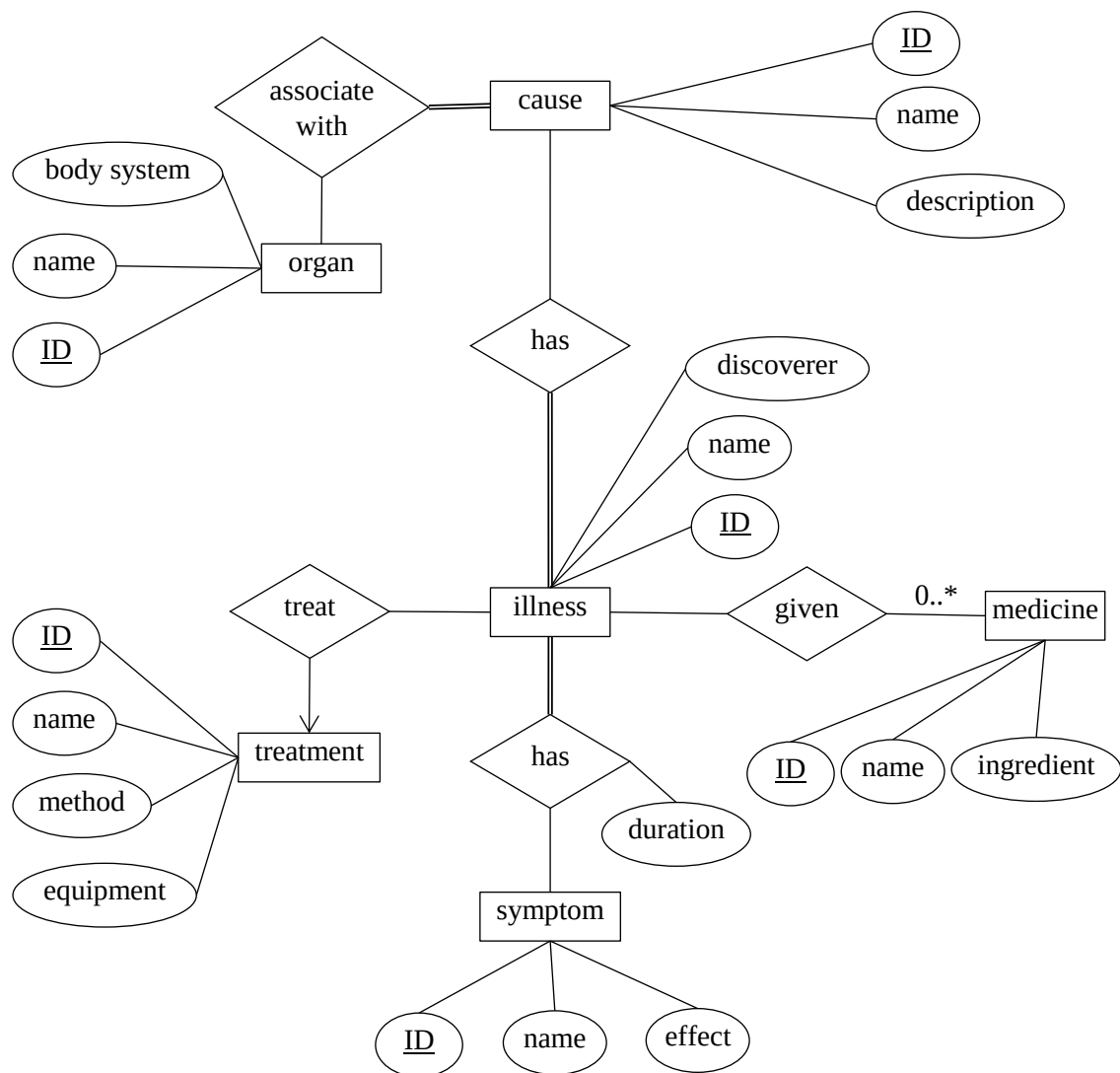
And b1.bid = b2.bid

- g) Can one reviewer rate a book multiple times? Why?
NO. When a reviewer rate a book with the same score on the same date as the first time, we can't distinguish the later tuple with the first tuple.

Q2. Suppose you are a medical school student. You want to design an ER diagram to model illnesses, treatments, and medicines under the following assumptions. (35 marks)

- Each **illness** is described as **ID, name, and discoverer's name**.
- Every **illness** is associated with one or multiple **cause(s)**.
- Each **cause** is described as **ID, name, and description**.
- Some causes are associated to one or multiple **organ(s)** to describe where the **causes** happen.
- Each **organ** is defined by **ID, name, and body system**.
- Every **illness** is associated with one or multiple **symptom(s)**. Each association also describes **how long** the **symptom** occurs.
- A **symptom** is defined by **ID, name, and effect**.
- Each **illness** is treated by at most one **treatment**.
- Each **treatment** is defined by **ID, name, equipment, and method**.
- Each **illness** is given zero, one, or multiple **medicine(s)**.
- If a **medicine** is given to an **illness**, the quantity is modeled.
- Each **medicine** is defined by **ID, name, and ingredient**.

You do not need to make more assumptions on your design.



Q3. Given the tables, what is the execution outcome (including both table title and table content) of the following queries? (4 marks for each)

<i>sID</i>	<i>sname</i>	<i>age</i>
101	Alice	19
102	Bob	18

Table 1 Student

<i>sID</i>	<i>course</i>	<i>score</i>
101	Database	80
102	Database	65

Table 2 Enroll

<i>sID</i>	<i>course</i>	<i>remark</i>
101	Database	Only a joke
102	Discrete Math	Excellent course

Table 3 Comment

- a) SELECT * FROM Enroll, Comment WHERE Enroll.course = Comment.course
- b) SELECT * FROM Student NATURAL JOIN Enroll NATURAL JOIN Comment
- c) SELECT * FROM Enroll NATURAL RIGHT OUTER JOIN Comment
- d) SELECT sID, course FROM Enroll JOIN Comment

a)

SID	COURSE	SCORE	SID	COURSE	REMARK
101	Database	80	101	Database	Only a joke
102	Database	65	101	Database	Only a joke

b)

SELECT * FROM Student NATURAL JOIN Enroll

SID	SNAME	AGE	COURSE	SCORE
101	Alice	19	Database	80
102	Bob	18	Database	65

SELECT * FROM Student NATURAL JOIN Enroll NATURAL JOIN Comment

SID	SNAME	AGE	COURSE	SCORE	REMARK
101	Alice	19	Database	80	Only a joke

c)

SCORE	SID	COURSE	REMARK
80	101	Database	Only a joke
UNKNOWN/NULL	102	Database	Excellent course

d)

error